

WIDE AREA NETWORKS

Task 1: Draw a simple diagram (on paper or using an online tool like draw.io) showing how your phone connects to Instagram's servers from your home WiFi, including the ISP and at least two Wide Area Network (WAN) devices (like routers or switches) in the path.

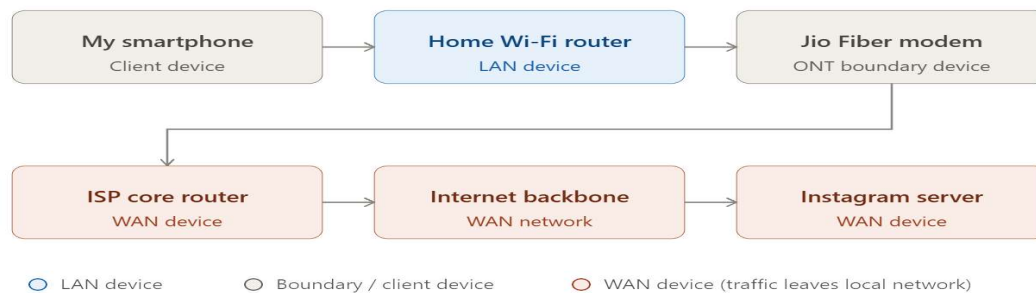


Diagram: Phone -> Home Wi-Fi Router -> Jio Fiber ONT/Modem -> ISP Core/Edge Router (WAN device) -> Internet Backbone (WAN) -> Instagram Data Center Server (WAN device). The two WAN devices in the path are the ISP Core/Edge Router and the Internet Backbone routers, since traffic leaves the local home LAN and travels across the wide area network to reach Instagram's servers.

Task 2: Create a table listing three types of WAN technologies (such as MPLS, Leased Line, and Broadband Internet) and compare them based on speed, cost, and where you might see them used in real life (e.g., Zomato office, IRCTC ticketing, your home).

WAN Technology	Typical Speed	Relative Cost	Real-Life Use Case
MPLS (Multiprotocol Label Switching)	Consistent 2 Mbps - 1 Gbps, low latency & jitter	High	Zomato/Flipkart HQ-to-branch links needing guaranteed performance and QoS for critical business apps
Leased Line	Dedicated, symmetric (e.g., 10-100 Mbps+), very reliable	High	IRCTC ticketing servers or bank data centers needing a dedicated, always-on, uncontended connection
Broadband Internet	Variable, shared bandwidth (e.g., 50 Mbps - 1 Gbps on Jio Fiber)	Low	Home use, small offices, or SD-WAN setups where cost matters more than guaranteed performance

Task 3: Research and write a short explanation of how a VPN (Virtual Private Network) uses WANs to let you access apps like Spotify or YouTube securely from a public WiFi at a coffee shop. Hint: Focus on how the VPN creates a private tunnel over the public WAN.

Spotify or YouTube, even when connected to an unsecured public WiFi network like one in a coffee shop.

When a device connects to public WiFi, its data still has to travel through the same physical path as any other internet traffic: the coffee shop's router, the ISP's WAN infrastructure, and the internet backbone, before reaching the destination server. A VPN does not change this physical path. Instead, it protects the data that travels along it.

When a VPN is activated, the device first establishes a connection to a VPN server. All data exchanged between the device and this VPN server is encrypted, meaning it is converted into a scrambled format that cannot be read by outsiders. This secure connection is commonly referred to as a "tunnel," because the data effectively travels through a private, protected channel even though it is physically passing over the same shared public network as other users.

Task 4: Imagine Flipkart wants to connect its warehouses in Ahmedabad, Mumbai, and Delhi. Suggest which WAN technology would be best for them and justify your answer with two reasons.

MPLS (Multiprotocol Label Switching) is a way networks (usually run by ISPs or telecom providers) speed up and control how data travels between locations.

Normally, every router along the way has to look at a packet's full destination address and figure out where to send it next — this takes a bit of time at every hop. MPLS speeds this up by attaching a small **label** to each packet as it enters the network. Every router after that just reads the label (not the full address) and quickly forwards it based on a simple lookup table, swapping the label at each hop until it reaches the destination.

MPLS would be the best choice for connecting Flipkart's Ahmedabad, Mumbai, and Delhi warehouses.

Reason 1: Speed and reliability of data flow

MPLS gives a fixed, reliable amount of bandwidth and very low delay, along with QoS (Quality of Service), which lets important traffic get priority. This matters for Flipkart because things like real-time inventory updates, order processing, and coordination between warehouses need to happen quickly and smoothly.

Reason 2: Stays online consistently

MPLS is very reliable and usually comes with backup paths built in, plus the provider promises a high uptime guarantee. This is important because if the connection between warehouses goes down, it can delay order fulfillment and disrupt stock updates — so the network needs to stay up almost all the time.